

CNN-Based Cell Segmentation in Multi-Modal Images

Project Description

Building on the original Ronneberger et al., (2019) architecture and the refinement by Falk et al., (2019) we propose a framework for segmenting individual cells in 2D fluorescence, differential interference contrast and phase-contrast microscopy using a modified U-Net convolutional network.

As part of the hackathon, we explored two approaches to training: a sliding window method and a random cropping strategy. The sliding window method systematically extracts overlapping patches from full images, while the random cropping strategy samples image regions randomly. Although the sliding window method ensures comprehensive coverage, random cropping was found to be significantly faster and more flexible during hyperparameter tuning.

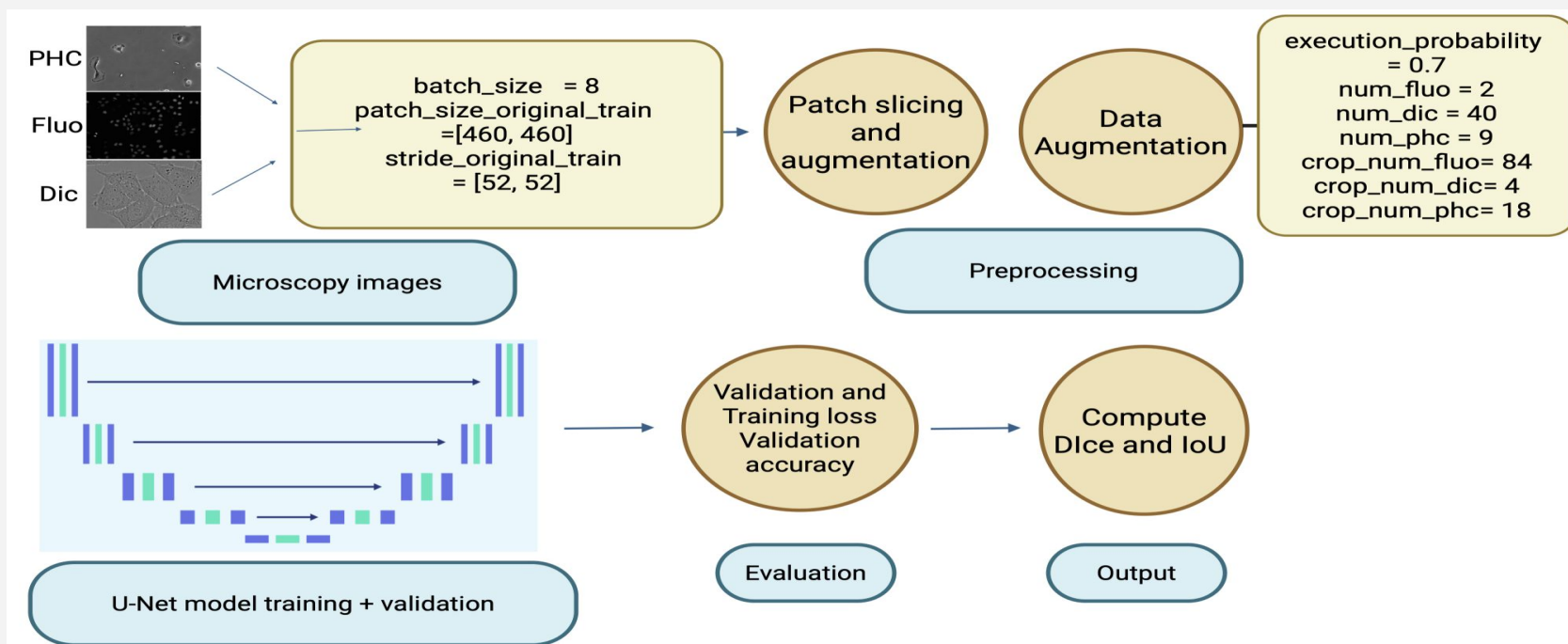


Fig. 1: Automated cell segmentation workflow.

Data & Methodology

The datasets are from the Cell Tracking Challenge (Ulman et al., 2017), consisting of 2D time-lapse images from three modalities:

- **Fluo-N2DL-HeLa:** Nuclei of transfected HeLa cells (resolution: 0.19×0.19 μm; 1100×700 pixels), with highly variable nuclear signal.
- **DIC-C2DH-HeLa:** HeLa cells (0.645×0.645 μm; 512×512), similar cell/background intensities, highly heterogeneous signal within cells.
- **PhC-C2DH-U373:** Glioblastoma astrocytoma U373 cells (0.65×0.65 μm; 696×520).

Training included 40 DIC, 9 PhC, and 2 fluorescence images. To enhance model generalization, training images underwent preprocessing (float conversion, normalization) and random augmentations (rotations, flips, elastic deformation, contrast/brightness changes).

These augmentations were applied with a specified execution probability to introduce variability and prevent overfitting. Finally, all images were converted to tensors. Training labels underwent corresponding spatial transformations (rotations, flips, and elastic deformation) to ensure alignment with the augmented images. Labels were also converted from instance to semantic format and transformed into tensors. For validation and test sets, only basic preprocessing steps – normalization and tensor conversion – were applied to ensure consistent evaluation without introducing randomness.

Evaluation metrics

For model evaluation the Dice coefficient, measuring overlap between predicted and true masks was used.

In terms of segmentation masks it can be seen as:

$$Dice = \frac{2 \sum_i p_i g_i}{\sum_i p_i + \sum_i g_i}$$

Here, p_i = predicted value (0 or 1, or probability) and g_i = ground truth value (0 or 1).

Further the IoU, quantifying the ratio of the intersection to the union of predicted vs. true pixel sets was used. IoU stands for Intersection over Union.

In pixel terms for segmentation masks it can be seen as:

$$IoU = \frac{\sum_i p_i g_i}{\sum_i p_i + \sum_i g_i - \sum_i p_i g_i}$$

where: p_i = predicted pixel and g_i = ground truth pixel.

The validation and training loss were determined using the mixed loss, defined as the $dice_ratio \cdot dice_loss + (1-dice_ratio) \cdot CrossEntropyLoss$.

The dice loss is: 1 - mean dice coefficient. The dice ratio was chosen to be 0.5 based on the model performance balancing sensitivity and specificity.

The validation accuracy was determined using TorchMetrics by dividing the amount of correctly predicted pixels through the amount total pixels.

1. **U-Net++** is an extension of the classic U-Net, featuring a dense mesh of connections between the encoder and the decoder. Using nested and stepped paths enables features from different resolution levels to be used more effectively.

2. **Attention U-Net** adds an attention gate mechanism to the skip-connection paths so the network can automatically focus on the most relevant areas (in this case, cells) while suppressing background noise.

3. **RRCNN U-Net** combines U-Net with recursive Residual Recurrent blocks.. Recursion allows features to be processed multiple times.

4. **RRCNN + Attention** includes the U-Net RRCNN with an attention mechanism so that the network processes features repeatedly and residually and also selects relevant segmentation areas.

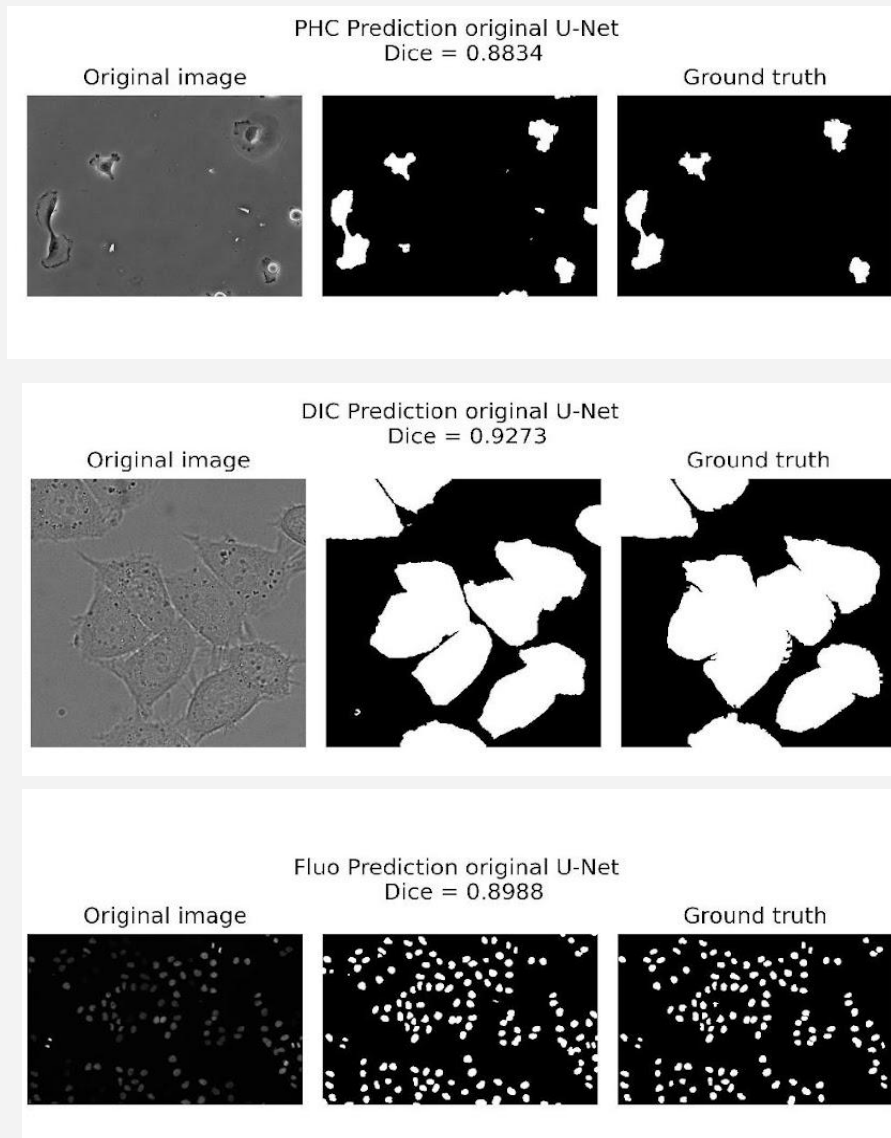
5. **BiFPN** replaces the standard decoder's simple upsampling and concatenation steps with learnable, attention-based fusion blocks. At each decoding level, the upsampled feature map and its corresponding encoder skip connection are aligned in the channel dimension.

Comparison of Different Model Architectures

Architecture	Best Loss	Layers	MIoU	Dice
U-Net	LogCoshDL	4-16-32-64	0.891	0.943
BiFPNU-Net	DiceLoss	4-16-32-64	0.90	0.926
AttU-Net	DiceLoss	4-16-32-64	0.890	0.914
RRU-Net	DiceLoss	4-16-32-64	0.890	0.907
AttRRU-Net	CombineLoss	4-8-32-64	0.883	0.901
U-Net++	DiceLoss	4-16-32-64	0.878	0.857

Table 1: To find the best hyperparameters, we ran a lot of experiments. To speed up the computations involved, we abandoned the sliding window method in subsequent steps, instead using random crops of the images. The results of the best models trained on the DIC dataset are presented for the whole test data.

Results : U-Net



When binarizing the U-Net's output at a probability threshold of 0.5, a high-quality contours across all three modalities were obtained (Fig. 1). In phase-contrast images, the model cleanly delineates cell bodies (Dice = 0.8834) with only minor “speckle” artifacts. In DIC, crisp edges yield near-perfect overlap (Dice = 0.9273). In fluorescence, the network successfully captures bright nuclei while suppressing background noise (Dice = 0.8988). The 0.5 cut-point therefore offers an excellent balance between sensitivity and precision and is used as our default for downstream analyses.

Fig. 2: Segmentation at a threshold of 0.5 for each image modality. Showing for each the original image, the prediction and the ground truth. The dice coefficient for the Phase contrast is 0.8834 (left), for the differential interference contrast is 0.9723 (middle) and for fluorescence microscopy 0.8988 (right).

Results : U-Net

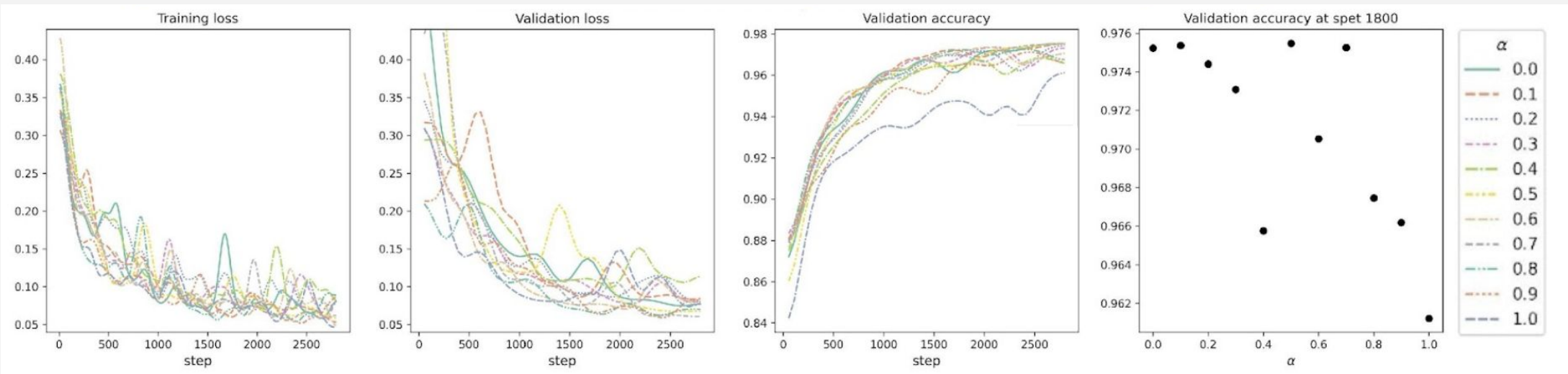


Fig. 3: Training and validation loss and accuracy for different α values. From left to right: Training loss, Validation loss, Validation accuracy, Validation accuracy at step 1800 and legend.

Effect of the α -parameter on training dynamics. Nine instances of our model with $\alpha \in \{0.0, 0.1, \dots, 1.0\}$ were trained. Training loss, Validation loss, and Validation accuracy converge rapidly to ≈ 0.05 loss and ≈ 0.97 accuracy for every α . The final validation accuracy at step 1800 varies by $< 1\%$ across α , with a slight optimum around $\alpha \approx 0.2$. In practice, performance is extremely robust to α in $[0, 1]$.

Model Performance Across Training and Testing Modalities

It can be seen that the model achieves high Dice and IoU scores when training and predicting the same modality and when using the combined modalities.

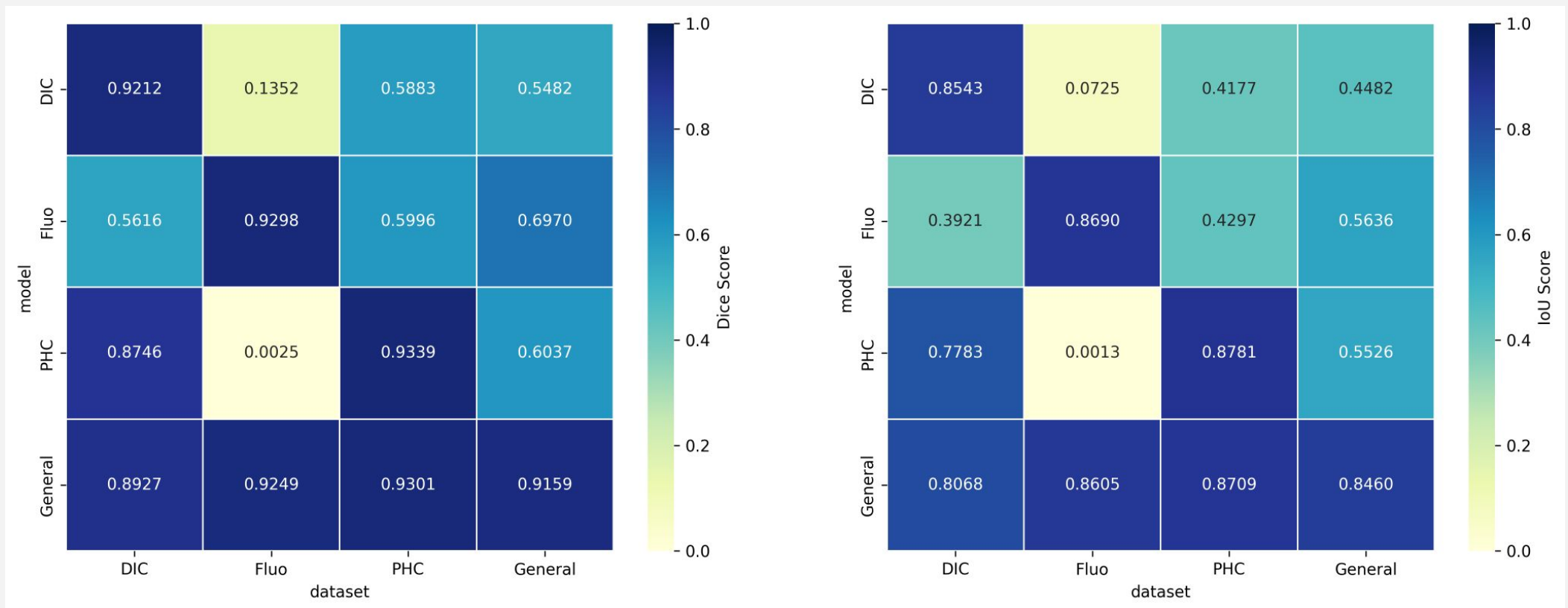


Fig. 4: The dice score (left) and IoU score (right) for the model trained on PHC, DIC or Fluo and combined dataset (general) are displayed. For the different modalities 5 images of each were used as a test set.

Comparison of Different U-Net Architectures & Losses

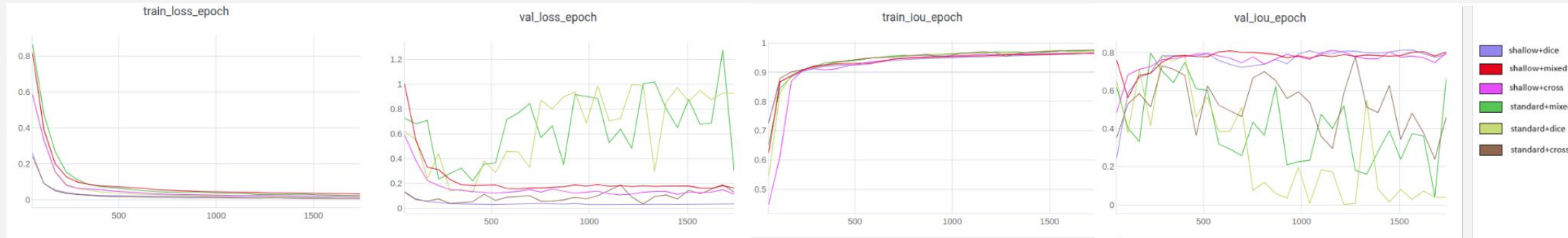


Fig. 5: Training and validation loss, train and validation IoU for different U-Net architectures and different losses

The main focus of the experiments was to test models with different weight losses. However, we also wanted to verify the influence of model architecture on model predictions. Models with a different number of layers and using different losses were trained. The first architecture was the 'standard' architecture, consisting of five layers with 64, 128, 256, 512 and 1,024 channels. The other model used only three of these layers with a channel count of 64, 128 and 256. The losses used for training were dice loss, cross-entropy loss, and a mixed loss — a combination of the previous two with equal weights.

Legend:

shallow – number of channels: [64, 128, 256]

standard – number of channels: [64, 128, 256, 512, 1024]

dice – trained only with dice loss

cross – trained only with cross loss

mixed – trained both with cross and dice loss (equal weights)

The results showed that even though the results were very similar during training, the shallow model significantly outperformed the standard model on the validation set. Not only did it achieve better scores using the IoU metric but was also more stable. The shallow model achieved great results for all of the chosen losses while the standard model performed best with the cross-entropy loss. The difference, however, was not very significant.

References

- Ronneberger O, Fischer P, Brox T. (2015)U-Net: Convolutional Networks for Biomedical Image Segmentation, MICCAI 2015, 234–241
- Falk T, Brox T, Ronneberger O et al. (2019)U-Net: deep learning for cell counting, detection, and morphometry, Nature Methods 16, 67-70
- Ulman V, Maška M, Harder N, Rohr K, Kozubek M, Ortiz-de-Solorzano C, et al. (2017)An objective comparison of cell-tracking algorithms, Nature Methods 14, 1141–1152
- Md Zahangir Alom, Tarek M Taha, Christopher Yakopcic, Stefan Westberg, Paheding Sidike, Mst Shamima Nasrin, Brian C Van Esesn, Abdul A S Awwal, and Vijayan K Asari, “The history began from alexnet: a comprehensive survey on deep learning approaches,”
- Mohammad Havaei, Axel Davy, David Warde-Farley, Antoine Biard, Aaron Courville, Yoshua Bengio, Chris Pal, Pierre-Marc Jodoin, and Hugo Larochelle, “Brain tumor segmentation with deep neural networks,” Medical image analysis, vol. 35, pp. 18–31, 2017.